



Theoretical and experimental analysis of precipitation and solubility effects in lithium-sulfur batteries

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ABSTRACT

A rate-dependent nucleation theory is developed to analyze the dependence of the capacity of lithium-sulfur (Li-S) batteries on discharge rates. The theory is based on expressing the nucleation rate of solid products in terms of the surface oversaturation of the precipitating species and rewriting the classical Kolmogorov-Avrami model in the form of a differential system of equations. By coupling this system of equations with the multicomponent transport equations we are able to explain qualitatively well the decrease of the specific capacity with increasing discharge rates and the characteristics of the discharge curves under variable discharge currents in Li-S batteries. Special attention is given to accurately determine the values of model parameters from experimental data, including the solubility constants of the solid products and the mobilities and diffusivities of the ions. The anodes of the batteries used in our analysis were made using Li foils and the cathodes were fabricated using freestanding carbon nanotube foams, on which solid sulfur was uniformly distributed to maximize the dissolution rate. The experimental data combined with the theoretical analysis show that the capacity limitation in our batteries is due to surface passivation by the final reaction products. In order to increase the specific capacity of Li-S batteries under normal discharge rates we conclude that it is better to avoid the solid precipitation of intermediate lithium polysulfides (LiPS) by using electrolytes with a high solubility of LiPS. In addition, it is better to use catalyst particles in the cathode that would promote the nucleation rate at specified locations. The number of catalyst particles should not be too small to avoid electrolyte oversaturation or too large to avoid the rapid passivation of the carbon surface with solid products.

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1. Introduction

Environmentally friendly, relatively inexpensive because of the low cost of sulfur, and having a high theoretical capacity, lithium-sulfur (Li-S) batteries are promising candidates to replace traditional Li-ion batteries in many applications from portable electronics to electric vehicle transportation and grid storage. Li-S batteries have a theoretical capacity of 1672 mAh/g_S and, when built using a Li metal anode, their specific energy density is 3–5 folds higher than that of Li-ion batteries [1–5]. Despite their large specific capacity, the current Li-S batteries with liquid electrolyte have a large electrolyte/sulfur (E/S) ratio, which decreases their

practical capacity much below the theoretical limit [6–10]. Designing Li-S batteries with low E/S ratios will require a detailed understanding of the physical and chemical processes that take place inside the cell during discharge and charge. This understating will likely come from a combination of experimental and theoretical investigations of the materials properties, precipitation, dissolution, and kinetic processes in the battery.

Although there is much experimental research on Li-S batteries the number of theoretical and numerical investigations is rather limited. Most of the existing theoretical models are based on the mathematical model proposed by Kumaresan et al. [11] and can describe fairly well the discharge voltage under low discharge currents. However, the existing models are somewhat contradictory and often use assumptions that are either far away from the experimentally measured data or not verified under high discharge rates. These models generally fail to describe the rate-dependent capacity of conventional Li-S batteries. In this article we combine

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theoretical and experimental data to develop a physics-based model for the discharge of Li-S batteries, which is valid under both low and high discharge rates. Our model is also based on the original transport model presented in Ref. [11] but carefully accounts for oversaturation, nucleation, precipitation, and surface coverage phenomena under variable discharge rates.

There is much debate in the literature about the reasons for the rate-dependence of the discharge capacity in Li-S batteries. We are currently aware of only a few simulation attempts to model the decrease of the discharge capacity of Li-S batteries with increasing discharge rate [12–17]. Zhang et al. [12,13] have attributed the decrease of the discharge capacity to the sluggish transport characteristics of anions and cations in the electrolyte, which results in a deficiency of sulfur ions in the cathode at high discharge rates. The electrolyte used in Ref. [12] is sulfolane-based which is possible to have a very low conductivity. To reproduce the voltage profiles at high discharge rates the authors decreased the mobility and diffusivity coefficients of the ions by 1-to-2 orders of magnitude below the values currently used for 1,2-dimethoxyethane (DME):1,3-dioxolane (DOL)-based electrolytes. In this way, the electrolyte conductivity becomes 1-to-2 orders of magnitude smaller than the values of the conductivity reported experimentally for DME:DOL (1:1 v:v). For instance, the ionic conductivity of Li_2S_6 calculated using the values of the mobility coefficients used in Ref. [12] is 0.068 S/m, which is more than one order of magnitude smaller than the experimental value of 1 S/m in DME:DOL (1:1 v:v). Although it is possible to design electrolytes with high viscosity, this model cannot explain the decrease of the capacity at high discharge rates in batteries with typical electrolytes, which have a conductivity of around 1 S/m. In addition, the model predicts a relatively fast recovery of the capacity after the rate of the discharge is decreased from 1 C to 0.1 C. The recovery time should be of the order of seconds according to the model parameters, while most of the experimental data shows that the recovery is much slower, of the order of hours.

Thangavel et al. [15] explain the decrease of the discharge capacity by a combination of two phenomena. First, they assume that the cathodes simulated in their analysis are made of porous particles with pore radius of 3 nm. At high discharge rates the entrance of these pores is being clogged with Li_2S , thus reducing the capacity of the battery. This model explains qualitatively well the rate-dependence of the total capacity, however it is applicable specifically to cathodes made of porous particles, such as the ones presented in their work. If one applies the model to cathodes made of nanoscale non-porous particles or CNT networks (like in this work), the model predicts that the discharge capacity does not change significantly with the discharge rate. Second, the authors assume that the Li_2S reaction product can grow on the surface of the particles up to a maximum tunneling thickness of 5 nm. This assumption is not supported by scanning electron microscopy (SEM) and atomic force microscopy (AFM) images taken by other authors, which show that the reaction product can grow in shapes as large as 500 nm [14,18] or even 1 μm and above [19–21]. In addition, the probability of electron tunneling across the potential barrier of the insulating layer is negligibly small if the thicknesses of the barrier is larger than 2 nm and the voltage drop across the barrier is below 1 V (see, for instance [22]), which implies that the carbon is passivated relatively fast, as soon as the Li_2S starts depositing on the surface of the carbon.

Hofmann et al. [16] use a slightly modified version of Kumaresan's model [11] and showed that they correctly reproduce the final capacities of the Li-S batteries if the consumption fraction of solid $\text{S}_{8(s)}$ is carefully considered. Their model considers four reactions at the cathode, polysulfide shuttle effects, and irreversible transport of active material from cathode to anode. Although the

model does not explain why the elemental sulfur is not fully dissolved and consumed during discharge, it still provides a simple way to model capacity loss at high discharge rates.

A model that recently received increased attention in the research community is to attribute the premature death of Li-S batteries to surface coverage with solid Li_2S [14,17,23]. To explain the dependence of capacity on discharge rate, Ren et al. [14] assume that the nucleation rate is rate-dependent according to an equation initially proposed by Milchev et al. [24] for the attachment rate in the electrolytic nucleation of metals and later used by Lau et al. [25] to study precipitation effects in Li-air batteries. However, the rate-dependent precipitation theory presented in Ref. [24] was developed for the electro-crystallization of metals, in which the ions in the electrolyte attach directly to the surface of the electrode. In the case of Li-S batteries, after the formation of the initial nuclei seeds, the Li_2S molecules diffuse in the electrolyte and precipitate on the surface of the existing seeds, away from the surface of the electrode. The growth of the solid Li_2S is phenomenologically different from the electrolytic nucleation in metals because Li_2S is highly insulating and does not contain free electrons to promote electrochemical deposition on its surface (like the metal-on-metal deposition would do). To account for the rate of surface coverage the authors use a two-dimensional version of the Kolmogorov phase transformation theory [26], which results into an integro-differential system of equations. By solving these equations self-consistently with the classical multicomponent transport equations, the authors show that the nucleation rate is faster at higher discharge rates, which leads to a faster surface coverage and, hence, a decreased capacity. The results provided by the model are also supported by SEM images of the cathode coverage and the model correctly explains the rate-dependent capacity in Li-S batteries.

In this article we develop a rate-dependent nucleation theory appropriate for the solid reaction products in Li-S batteries and use this theory to explain qualitatively the capacity loss at high discharge rates. The theory is similar to the model developed by Ren et al. [14] in the sense that it attributes that death of Li-S batteries to surface passivation, however, in this work, the mathematical equations are framed entirely in the form of a system of partial differential equations that can be implemented easily in ordinary or partial differential equation solvers. The equation for the nucleation rate is relatively similar to the one used in Ref. [14] but, here, it is derived on a phenomenological basis, specifically for the precipitation of reaction products in Li-S batteries. In this way, the derivation for the nucleation rate can be viewed as proof for the equation employed in Ref. [14].

Finally, one should note that finite-element models for the simulation of Li-S batteries have also been developed by other authors. In general, these models describe fairly well the voltage as a function of capacity for low discharge or charge rates but do not predict correctly or have not been tested yet for higher rates. For instance, the model of Kumaresan et al. [11] predicts that the discharge capacity is independent on the discharge rate for rates below 1 C. Similarly, the models presented in Refs. [27,28] cannot explain the decrease of the specific capacity when the discharge rate is increasing to 1 C and beyond.

2. Experimental

The cells used in this study were made using Li foil anodes (0.25 mm, MTI), polypropylene separators (25 μm , Celgard 2400), and carbon nanotube (CNT) foam-based cathodes. The CNT foams contained CNTs with an average diameter of 10 nm and a length of 2 mm, which were assembled in a freestanding three-dimensional mesh using the method presented in Refs. [7,29]. The obtained CNT foams had a thickness of 100 μm and a density of 125 mg/cm^3 . The

porosity and specific area of the cathodes (without sulfur) were measured using the nitrogen adsorption-desorption method and were estimated at 96% and 392 m²/g, respectively. An SEM image of the freestanding CNT mesh is presented in Fig. 1a. Solid sulfur microparticles were added to the cathodes and heated at 158 °C to form a uniformly distributed layer of S_{8(s)} as shown in the SEM image presented in Fig. 1b. The sulfur content of the cathodes is 60% and the sulfur loading per electrode is 1.9 mg/cm². The sulfur content was estimated based on the initial amount of sulfur to be approximately 9.5% by volume. The electrolyte was composed of 1 M bis(trifluoromethane)sulfonimide (LiTFSI) and 0.1 M LiNO₃ in DME:DOL (1:1 v:v). Finally, the anodes, the separators, and the cathodes were assembled into CR2032-type coin cells.

The voltage characteristics of the cells are represented in Fig. 2 for different discharge rates ranging from 0.1 C to 1 C. As expected, the specific capacity of Li-S batteries is rate-dependent. The capacity of the first plateau is relatively independent on the discharge rate, while the capacity of the second plateau decreases with increasing discharge rate. An important feature of the discharge curves of our batteries is that they have two dips: one at beginning of the discharge (“A”) and one at approximately 400 mAh/g_s (“B”). Dip “A” is due to the fact that, in our batteries, the active area of the cathode is relatively small at the beginning of the discharge (see Fig. 1b that shows that the initial solid S_{8(s)} covers

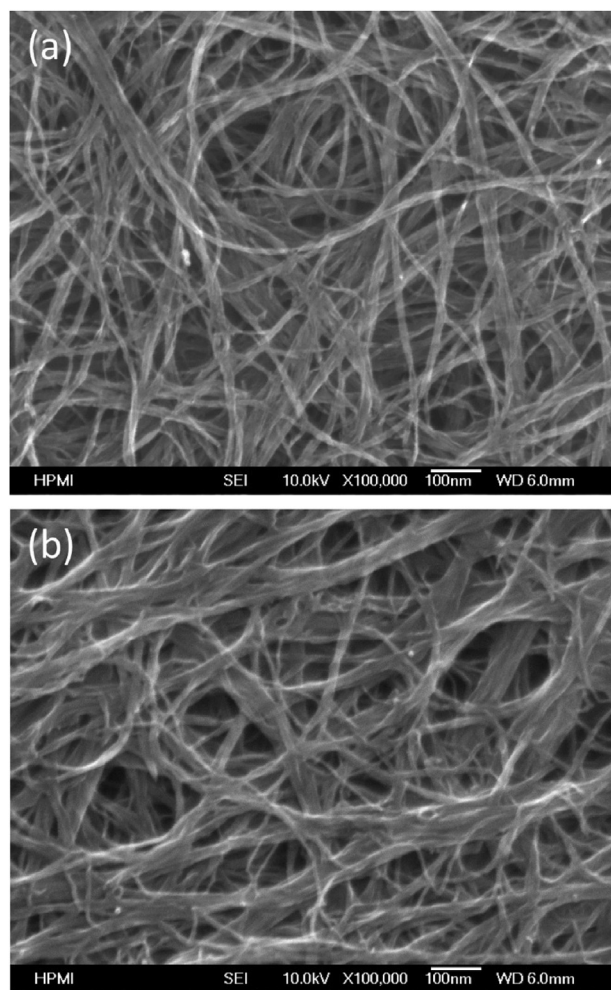


Fig. 1. SEM image of the CNT cathode before loading (a) and after loading (b) with solid sulfur, S_{8(s)}. The solid S_{8(s)} is deposited approximately uniformly as a thin layer on the surface of the CNT bundles.

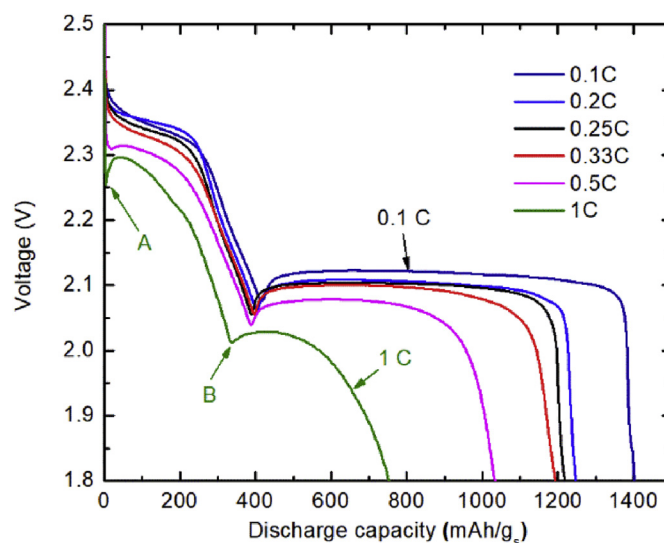


Fig. 2. Experimental discharge curves measured at different C rates.

practically the entire active area of the cathode). For this reason, the cells require a large initial overpotential to start the discharge of the battery. This phenomenon is particularly visible at high discharge rates and, in this study, will be modelled mathematically by a phenomenological equation, eq. (34). A similar behavior was observed in Refs. [17,30]. The voltage decrease at point “B” is due to electrolyte oversaturation, which is required to start the nucleation process.

To verify if the decrease of the discharge capacity with increasing discharge rates is due to the sluggish transport of the electrolyte in the cathode, we have repeated the experimental measurements presented in Ref. [13]. In particular, we have discharged the battery at a relatively “low” discharge rate of 0.2 C up to approximately 50%, after which we continued the discharge at a “high” discharge rate of 1 C. When the voltage of the battery reached 1.8 V (i.e. the battery is almost “dead”) we changed the rate to 0.2 C and measured the discharge voltage. The results of these measurements are represented in Fig. 3. Two important

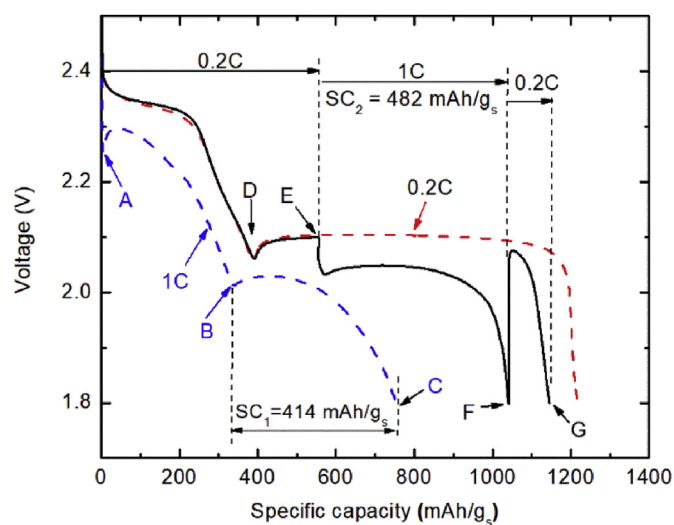


Fig. 3. Cell voltage under variable discharge rate of consecutive 0.2 C-1 C-0.2 C steps (continuous line). The dash lines represent the cell voltage under constant discharge rates of 0.2 C and 1 C, respectively.

observations can be made from these measurements.

1. Unlike in Ref. [13], the specific capacity at the end of the discharge with variable rate (see the continuous line in Fig. 2) is different from the discharge capacity obtained under the “low” discharge rate of 0.2 C. Although we have repeated these measurements for multiple times using different “high” and “low” discharge rate and switched from “low”-to-“high” and “high”-to-“low” rates at different states-of-discharge, the discharge capacity under variable discharge rate was always lower than the discharge capacity under the “low” discharge rate. This fact shows that the decrease of the discharge capacity with increasing discharge rates in our batteries is not due to the sluggish transport characteristics of the electrolyte as explained in Refs. [12,13]. The reason for the decrease of the specific capacity at high discharge rates should be related to the different cathode passivation dynamics under “low” and “high” discharge rates.
2. The specific capacity measured at 1 C from the moment when the solid product deposition starts until the death of the battery is significantly lower than the specific capacity obtain at the same rate during the variable discharge measurement (compare $SC_1 = 414 \text{ mAh/g}_S$ to $SC_2 = 482 \text{ mAh/g}_S$ in Fig. 3). This fact is somewhat counterintuitive as we expect some of the active area of the carbon to be passivated with solid products between points “D” and “E” and reduce the active area available for discharge from point “E” to “F”. This phenomenon is due to the different dynamics in which nuclei appear on the surface of the carbon, as it will be shown in the results section.

Both observations will be explained in section 4 by using the mathematical model developed below.

3. Theory

To develop a mathematical model for the discharge of Li-S batteries we need to determine the fundamental transport and electrochemical processes that limit the discharge of these batteries. Identifying the main phenomena that take place during discharge will enable us to build a realistic model that describes the operation of the battery. The material parameters that enter into the governing equations need to be determined using data taken as much as possible from direct measurements of the material properties instead of fitting the final discharge curves with simulation results. The model developed below is based on the classical multicomponent transport equations in porous media, which can accurately capture ohmic and diffusion losses in the electrolyte. First, we focus on the role of precipitation and surface coverage in the cathode and then we summarize the model equations. Phenomena such as polysulfide shuttling and side reactions that take place mostly during charging are neglected in this work.

The anode is made of a Li metal. The only electrochemical reaction at the anode is assumed to be



The cathode is made of a matrix of carbon nanotubes (CNT) that are randomly distributed (see Fig. 1). Following the model by Kumaresan et al. [11], at the cathode, we consider the following electrochemical reactions



and dissolution/precipitation reactions



Although some authors consider that some of the electrochemical reactions (2)–(12) might not take place in Li-S batteries [31,32], due to the lack of conclusive data we include all of them in our analysis. Particular cases such as the case when lithium polysulfide (LiPS) is reduced from a higher order product to a lower order product by bypassing some of the intermediate reaction steps can be described in our model by increasing the reaction rates and solubilities of one or more of the intermediate products. In addition to the above reactions, it is likely that multiple disproportionation and possibly comproportionation reactions also take place during discharge [33,34]. These reactions can take place throughout the volume of the electrolyte and not necessarily at the CNT/electrolyte interface. Including these reactions in the model is relatively straightforward, however, this requires identifying the exact reactant and product species and determining the reactions rate coefficients carefully. The mass transport of the intermediate species can be described in the same way as presented in section 3.2 below. Although in this work we neglect disproportionation and comproportionation reactions, it is expected that these reactions play an important role during the charge and discharge and will provide a better understanding of the operation of Li-S batteries in the future.

3.1. Nucleation, precipitation, and surface coverage

The precipitation of solid products is believed to be the main reason for the active surface coverage and passivation. The model for nucleation, precipitation, and surface coverage developed in this section is based on the following observations:

1. Since the electric resistivity of Li_2S is large and the electron tunneling current is negligibly small for the thicknesses of Li_2S layers observed experimentally, electrochemical reactions can take place only at the electrolyte/carbon interface and not at the Li_2S /electrolyte interface. The carbon surface underneath the

Li_2S deposit product becomes passivated as soon as the Li_2S seeds are formed.

- The SEM images presented in the literature suggest that the size of the final Li_2S product can be as large as hundreds of nanometers or even micrometers. Therefore, we conclude that the S^{2-} ions (or Li_2S molecules in case such molecules are formed directly as a result of the cathode reaction) diffuse away from the carbon surface and eventually attach to existing Li_2S seeds, like in Fig. 4a and Fig. 4b.
- In agreement with the classical theory of precipitation, we assume that the Li_2S nuclei can form only on the surface of the carbon, as shown in Fig. 4c and d. Therefore, we exclude isolated formation of the Li_2S seeds away from the carbon/electrolyte interface. Such formations would be energetically less favorable than on the surface of the carbon. This fact is also suggested by current SEM studies that show that the Li_2S deposits on the surface of the carbon.

Although the above observations were made for the precipitation of Li_2S , we assume that they are also valid for any other precipitating species in reactions (8)–(12).

The driving force of the initial nucleation process (i.e. the formation of the solid seeds) is the oversaturation at the surface of the carbon, also called surface oversaturation, $S_{k,\text{surf}}$, where k denotes the solid reaction products $\text{Li}_2\text{S}_{8(s)}$, $\text{Li}_2\text{S}_{6(s)}$, $\text{Li}_2\text{S}_{4(s)}$, $\text{Li}_2\text{S}_{2(s)}$, and $\text{Li}_2\text{S}_{(s)}$. According to the classical nucleation theory, for

oversaturations $S_{k,\text{surf}} \gg 1$, the nucleation rate is proportional to $S_{k,\text{surf}}^N \exp(-B \ln^2 S_{k,\text{surf}})$, where N is a power coefficient that depends on the type of nucleation and B is a dimensionless thermodynamic parameter [35]. In our work, in order to simplify the numerical implementation and decrease the number of fitting parameters, we approximate the increasing exponential factor from the last expression with a power function. In this way, the number of nuclei of solid species k that appear per unit area and unit time is approximated by a power law

$$n_k = M_k (S_{k,\text{surf}}^N - 1) \quad (13)$$

where M_k and N_k are some fitting parameters. The “-1” term in eq. (13) was introduced to correctly describe the equilibrium condition under which $n_k = 1$ when $S_{k,\text{surf}} = 1$. It is important to notice that eq. (13) is accurate for $S_{k,\text{surf}} \gg 1$ and might need to be modified when modeling dissolution processes (such as the dissolution of $\text{S}_{8(s)}$). For this reason index k in (13) does not include $\text{S}_{8(s)}$.

The surface oversaturation is defined as $S_{k,\text{surf}} = \prod_i a_{i,\text{surf}}^{\gamma_{i,k}} / K_{sp,k}$, where $a_{i,\text{surf}}$ are the activities of the ionic species at the surface of the carbon, $K_{sp,k}$ is the solubility product, and $\gamma_{i,k}$ is the number of moles of ionic species i in the solid species k , which can be found by writing chemical reactions (7)–(12) in the form $\sum_i \gamma_{i,k} M_i \leftrightarrow M_{k(s)}$. Expressing activities as concentrations,

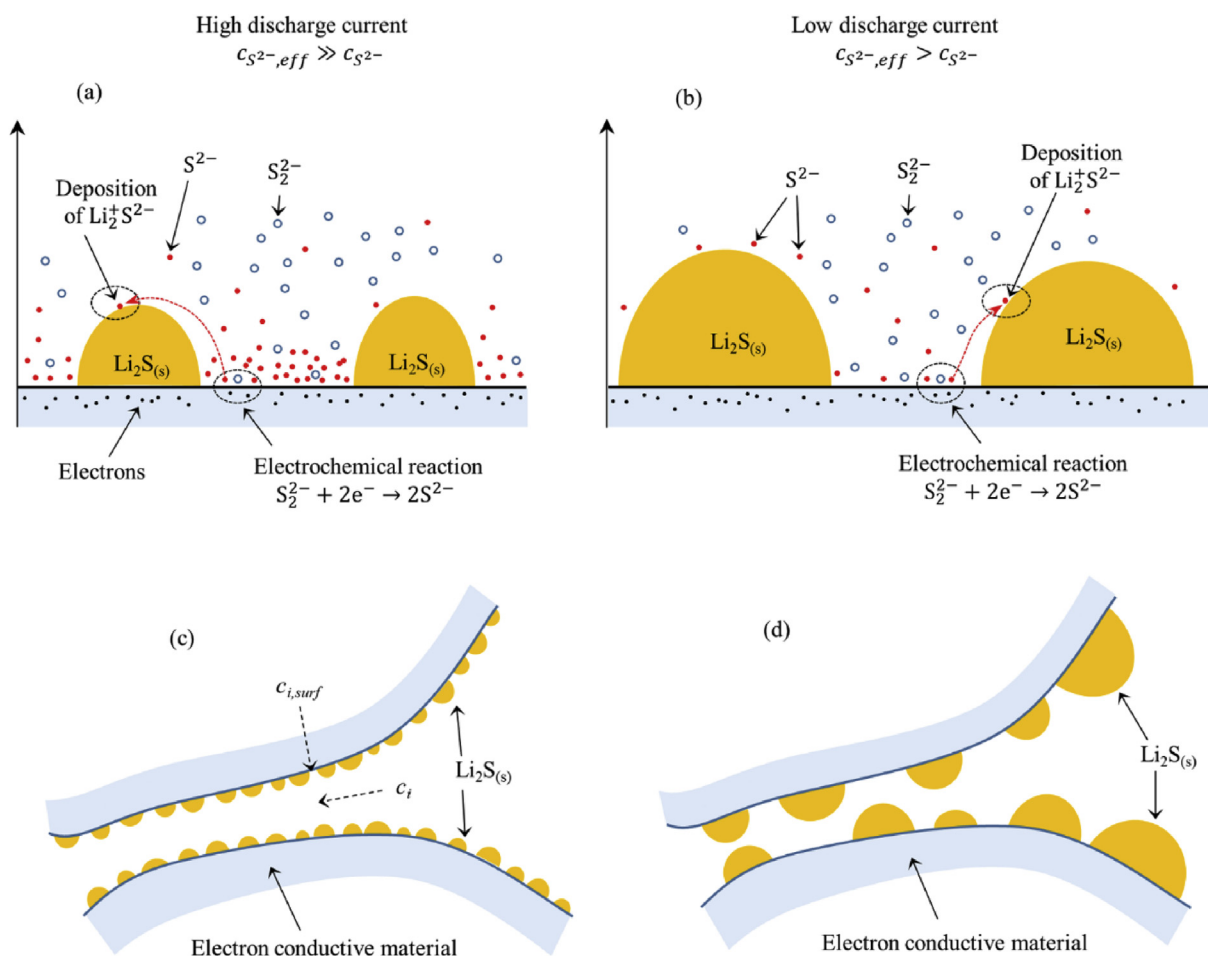


Fig. 4. Nucleation and growth of Li_2S in a micropore on the surface of the carbon. Li^+ ions are not represented to simplify the illustration. The concentration of S^{2-} ions on the surface of the carbon at high discharge rates (figs. (a) and (c)) is higher than the concentration on the surface at low discharge rates (figs. (b) and (d)). Therefore, the nucleation rate of Li_2S is higher at high discharge rates. The dash arrows in (c) denote the locations where concentrations $c_{i,\text{surf}}$ and c_i are defined.

we approximate

$$S_{k,surf} = \frac{\prod_i c_{i,surf}^{\gamma_{i,k}}}{K_{sp,k}} \quad (14)$$

where i denotes S_8^{2-} , S_6^{2-} , S_4^{2-} , S_2^{2-} , and S^{2-} . Notice that the concentrations of ionic species that enter in the definition of the surface oversaturation are equal to the concentrations of these species near the surface of the carbon, $c_{i,surf}$ (see the red dots in Fig. 4a and b). Since $c_{i,surf}$ increases with the discharge rate, the model developed below predicts a rate-dependent nucleation rate in Li-S batteries. This result is also consistent with experimental observations by other groups [14,36,37].

To compute the concentration of the ionic species at the surface the carbon ($c_{i,surf}$) we notice that the production rate of ionic species at the surface can be written approximately as $R_i \approx D_i \frac{c_{i,surf} - c_i}{L_i}$, where c_i is the concentration of this species away from the surface, D_i is the diffusivity of the ionic species, and L_i is a characteristic length describing the diffusion of species i in bulk electrolyte, but close to the carbon surface. In writing the last equation we assumed that the migration (drift) flux is much smaller than the diffusion flux, which is an approximation often used in Li-S systems and which allows to simplify the mathematical model significantly [15,16]. Notice that if the diffusing species are uncharged molecules such as Li_2S , the migration flux vanishes and the above approximation is accurate. The last equation shows that $c_{i,surf}$ increases with the production rate of species i at the carbon/electrolyte interface as $c_{i,surf} = c_i + \lambda_i R_i$, where $\lambda_i \sim L_i/D_i$ is a parameter that depends on the diffusivity and diffusion length of the ionic species next to the carbon surface. Due to the rough approximations used to derive this equation, the equation should be regarded as phenomenological and parameter λ_i should be treated as a fitting parameter. With these approximations, the oversaturation at the surface of the carbon can be written as:

$$S_{k,surf} = \frac{\prod_i (c_i + \lambda_i R_i)^{\gamma_{i,k}}}{K_{sp,k}} \quad (15)$$

Eq. (15) shows that the surface oversaturation increases with the reaction rate. This equation should be used with care in numerical simulations. It works relatively well for the case when there is a net generation of species i on the surface of the carbon (i.e. $R_i > 0$), however, it might create numerical instabilities if $c_i + \lambda_i R_i < 0$. Therefore, if this equation is used for other species than the generation of S^{2-} , the term in parenthesis should be set to 0 if $c_i + \lambda_i R_i < 0$ (notice that for species S^{2-} $c_i + \lambda_i R_i$ is always positive).

Eqs. (13) and (15) describe the nucleation rate in our model. It can be shown that the nucleation rate given by eq. (13) is mathematically similar to the nucleation rate used by Ren et al. [14] and Lau et al. [25] (see eq. (4) from these references), which applies to discharge at moderate and high rates. Indeed, in the case of large surface oversaturations, $c_{i,surf} \gg c_i$ and eqs. (13) and (15) imply that $n_k \approx \frac{M_k}{K_{sp,k}^{N_k}} \prod_i (c_i + \lambda_i R_i)^{N_k \gamma_{i,k}} \approx \frac{M_k}{K_{sp,k}^{N_k}} \prod_i (\lambda_i R_i)^{N_k \gamma_{i,k}}$. Since the reaction rates are proportional to $\exp\left(\frac{\alpha \eta}{V_T}\right)$ where α is the symmetry factor, V_T is the thermal voltage, and η is the overpotential (see eq. (23)), it follows that $n_k \sim \exp\left(\frac{\alpha \eta}{V_T}\right)$, where the proportionality factor is concentration dependent, which is the same as eq. (4) from Refs. [14] and [25].

Since precipitation is believed to play a major role on the

performance of Li-S batteries, one of the most important contributions of this work is to develop a physics-based model for surface coverage. Next, we denote the relative surface coverage of precipitate k by θ_k , where $0 \leq \theta_k \leq 1$. We make the assumption that the different types of precipitates are isolated from each other, so the total (relative) surface coverage is $\theta = \sum_k \theta_k \leq 1$. There exist many

mathematical models to describe θ in the literature. A common approach is to assume that the coverage is proportional to $(\varepsilon/\varepsilon_{initial})^\xi$, where ε and $\varepsilon_{initial}$ are the actual and initial volume fractions of the electrolyte and $\xi \sim 1.5$ is an empirical parameter describing the morphology of the precipitate. Although this model is widely used in the literature it does not properly take into consideration the nucleation rates of each precipitate. A more accurate model is to keep track of the size distribution of the nuclei at each location inside the cell and adjust it accordingly during the discharge of the battery [14,25]. This approach is equivalent to integrating over the number of nuclei at each moment in time. The model developed in Appendix A and summarized below is conceptually similar to the above integral model, however it is written in a purely differential form suitable for implementation in standard simulators. It is based on writing the Kolmogorov equation in differential form (instead of writing it in integral form like in Refs. [14,25]) and using the oversaturation in the transport channels (S_k) as the driving force for nucleation

$$S_k = \frac{\prod_i c_i^{\gamma_{i,k}}}{K_{sp,k}} \quad (16)$$

where i denotes S_8^{2-} , S_6^{2-} , S_4^{2-} , S_2^{2-} , and S^{2-} . The concentrations of the ionic species that enter in the definition of the oversaturation are equal to the concentrations of these species away from the carbon surface, c_i . As shown in Fig. 4a, due to the relatively large size of the deposit product, the concentration of the species in the channel (c_i) is different from the concentration of the species at the carbon surface ($c_{i,surf}$).

According to the discussion in Appendix A, the relative coverage with solid product k is described by the following system of equations

$$\frac{\partial \theta_k}{\partial t} = K'_k S_k (1 - \theta) F_k \quad (17)$$

$$\frac{\partial F_k}{\partial t} = S_k G_k \quad (18)$$

$$\frac{\partial G_k}{\partial t} = M_k (S_k^{N_k} - 1) \quad (19)$$

where K'_k are rate constants describing how fast the reaction product grows once a nucleation seed is formed, and F_k and G_k are integration functions that satisfy the following initial conditions $F_k(0) = G_k(0) = 0$. Notice that we introduced partial derivatives in eqs. (17)–(19) because, unlike in Appendix A where the derivation was performed at a given location inside the cathode, functions θ_k , F_k , and G_k now depend on time and position. Function G_k denotes the number of nuclei per unit area of species k at time t , while F_k measures the rate at which the surface coverage of one non-overlapping nucleus increases in time.

The total (relative) coverage of the solid products is given by $\theta = \sum_k \theta_k$. The surface coverage can be found by solving system (17)–(19) self-consistently with the transport system of equations presented in the next section.

3.2. Summary of model equations

Charge and mass transport are governed by the continuity equations for each species in the electrolyte and electrons in the CNT matrix

$$\frac{\partial \varepsilon_i}{\partial t} + \nabla \cdot (-z_i u_i' c_i \nabla \phi - D_i' \nabla c_i) + a(1 - \theta) R_i - \sum_{k'} \gamma_{i,k'} r_{k'} = 0 \quad (20)$$

$$\nabla \cdot (\sigma' \nabla V) + \sum_j R_j = 0 \quad (21)$$

where $R_i = \sum_j s_{ij} R_j$, z_i is the charge number of species i , c_i is the concentration of species i , a is the specific area computed by neglecting the solid precipitates, ϕ is the potential of the electrolyte, V is the potential of the electrons, s_{ij} are the stoichiometric coefficients of species i in electrochemical reaction j , R_j is the reaction rate in reaction j , and $r_{k'}$ is the consumption rate of species k' in the precipitation reaction. The stoichiometric coefficients of species i in electrochemical reaction j , s_{ij} , are obtained by writing electrochemical reactions (1)–(6) as $\sum_i s_{ij} M_i^{z_i} = e^-$. The (effective) mobilities and diffusivities in the porous material are related to the “bulk” mobilities and diffusivities using Bruggeman relations $u_i' = \varepsilon^b u_i$ and $D_i' = \varepsilon^b D_i$. Similar relation is used to compute the effective conductivity of the CNT network $\sigma' = \varepsilon_{CNT}^b \sigma$, where ε_{CNT} is the fraction of the CNT network.

The volume fraction of each solid reaction product changes in time because of precipitation/dissolution reactions

$$\frac{\partial \varepsilon_{k'}}{\partial t} = V_{k'} r_{k'} \quad (22)$$

where $V_{k'}$ are the partial molar volumes of the precipitates. The fraction of the electrolyte is computed as $\varepsilon = 1 - \varepsilon_{CNT} - \sum_{k'} \varepsilon_{k'}$.

Index i in eqs. (20)–(22) denotes the 8 mobile species in the electrolyte Li^+ , S_8 , S_8^{2-} , S_6^{2-} , S_4^{2-} , S_2^{2-} , and A^- (where A^- denotes the anions in the fresh electrolyte); j denotes the 5 electrochemical reactions for the formation of S_8^{2-} , S_6^{2-} , S_4^{2-} , S_2^{2-} , and S^{2-} (i.e. eqs. (2)–(6)); and k' denotes the 6 solid species $\text{S}_{8(s)}$, $\text{Li}_2\text{S}_{8(s)}$, $\text{Li}_2\text{S}_{6(s)}$, $\text{Li}_2\text{S}_{4(s)}$, $\text{Li}_2\text{S}_{2(s)}$, and $\text{Li}_2\text{S}_{(s)}$. Although the fresh electrolyte used in our experiments contained two different anions (TFSI^- and NO_3^-), to simplify the equations, we consider that both anions have the same mobility and diffusivity and describe them by A^- .

The reaction rates are related to the local concentrations of each species

$$R_j = \frac{i_{0j}}{F} \left[\prod_{i, s_{ij} > 0} \left(\frac{c_i}{c_i^{\text{ref}}} \right)^{s_{ij}} e^{\frac{\alpha \eta}{V_T}} - \prod_{i, s_{ij} < 0} \left(\frac{c_i}{c_i^{\text{ref}}} \right)^{-s_{ij}} e^{-\frac{(1-\alpha)\eta}{V_T}} \right] \quad (23)$$

$$r_k = k_{k'} \varepsilon_{k'} \left[\prod_i (c_i)^{\gamma_{i,k'}} - K_{sp,k'} \right] \quad (24)$$

where $\eta = V - \phi - U_0$ and c_i^{ref} are reference concentrations determined from the reference voltages of each electrochemical reaction $U_{j,\text{ref}} = U_0 + \frac{V_T}{n_j} \sum_i s_{ij} \ln \frac{c_i^{\text{ref}}}{1000}$. The reference concentrations are assumed to be equal to the initial concentrations so that $R_j = 0$ for a fresh battery, in which the open circuit voltage is $U_0 = 2.45\text{V}$. An

approximation that we made in eq. (23) is that we did not use the species concentrations at the surface of the carbon, $c_{i,\text{surf}}$ but, instead, we used concentrations c_i , which are defined away from the surface. This approximation is good because the overpotential is calculated in the terms of potential ϕ , which is also defined away from the surface of the carbon (and not at the carbon/electrolyte interface).

The reaction rate at the anode is

$$R_A = \frac{i_{0,\text{Li}^+}}{F} \left[\frac{c_{\text{Li}}}{c_{\text{Li}}^{\text{ref}}} e^{\frac{\alpha \eta_A}{V_T}} - e^{-\frac{(1-\alpha)\eta_A}{V_T}} \right] \quad (25)$$

where F is the Faraday's constant and $\eta_A = V_A - \phi$ is the overpotential in the anode reaction. Eq. (25) enters as a boundary condition in the transport system of equations. In the last equation V_A is the potential of the electrons in the Li metal. The voltage of the battery is computed as the potential of the cathode at the electric contact minus the potential of the anode.

If all the precipitation reactions are considered, eqs. (17)–(22) form a system of 30 differential equations that should be subject to boundary conditions and solved to compute variables c_i , $\varepsilon_{k'}$, θ_k , F_k , and G_k . Despite the large number of equations in (17)–(22) the system can be written in matrix form and implemented numerically easily because of its repetitive form. If only one species is considered to precipitate (e.g. Li_2S) the number of differential equations decreases to 14.

In the next subsections we focus on the determination of model parameters, including ion mobilities and diffusivities, solubility products, exchange current densities, reactions potentials, and nucleation-related parameters.

3.3. Ion mobilities and diffusivities

Ion mobilities, u_i , and diffusivities, D_i , are the most important parameters that describe mass and charge transport in bulk electrolytes. Unfortunately, the mobilities and diffusivities of the ions are difficult to measure directly and these parameters need to be determined indirectly from measurements of the electrolyte conductivity (κ) and transference numbers (t_i), which can be written as

$$\kappa = F \sum_i z_i^2 u_i c_i \quad (26)$$

$$t_i = \frac{z_i^2 u_i c_i}{\sum_i z_i^2 u_i c_i} \quad (27)$$

If the electrolyte conductivity and transference numbers of each ion are measured experimentally, eqs. (26) and (27) can be used to compute ions mobilities $u_i = \kappa t_i / z_i^2 F c_i$. In this work we use the experimental data by Fan et al. [38] and Suo et al. [39], which is in agreement with data published by other research groups [40,41]. This data is valid for LiTFSI dissolved in DME:DOL (1:1 v:v), which is the same electrolyte as in our experiments. To derive an analytical equation for the electrolyte conductivity we fit the experimental data with $\kappa(c_{\text{Li}}) = C_1 c_{\text{Li}} \times e^{-C_2/c_{\text{Li}}}$, where C_1 and C_2 are fitting parameters. This function was selected in order for the electrolyte conductivity to be in agreement with the dilute solution theory (in particular, κ should satisfy $\kappa(0) = 0$ and increase linearly with c_{Li} at low lithium concentrations); in addition, this functional form shows a decrease of the electrolyte conductivity at high lithium concentrations in agreement with the existing experimental data. The transference number of Li ions is fitted to function $t_{\text{Li}^+}(c_{\text{Li}}) = t_+ + t_- \tanh \frac{c_{\text{Li}} - C_3}{C_4}$, where C_3 , C_4 , C_5 , and C_6 are fitting

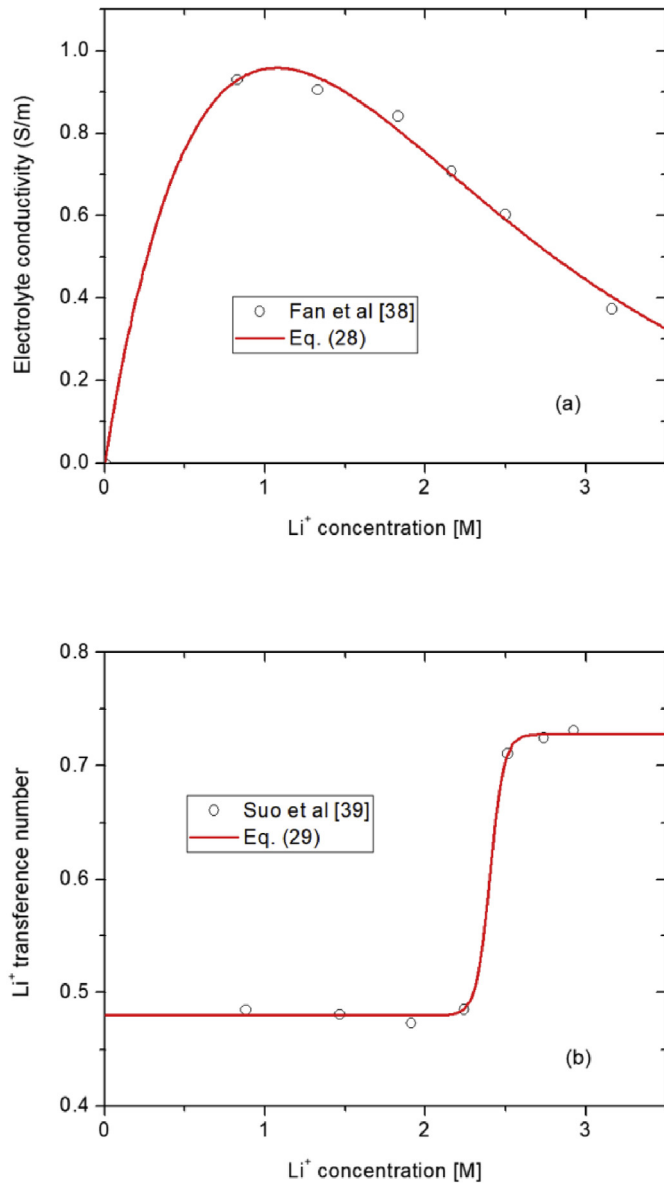


Fig. 5. Electrolyte conductivity and Li-ion transference. The horizontal axis of the experimental data was adjusted to indicate the Li ion concentration in the electrolyte.

parameters. The fitted and experimental data is presented in Fig. 5, where

$$\kappa(c_{Li}) = 2.43 \times 10^{-3} c_{Li} e^{-\frac{c_{Li}}{1074}} \quad (28)$$

$$t_+(c_{Li}) = 0.604 + 0.124 \tanh \frac{c_{Li} - 2,404}{87.3} \quad (29)$$

where the conductivity is measured in S/m and the lithium concentration in mol/m³. Using equations (26)–(29) we compute the mobility of Li ions (measured in m²/Vs) as

$$u_{Li} = 3.123 \times 10^{-9} e^{-\frac{c_{Li}}{1074}} \left[4.87 + \tanh \left(\frac{c_{Li} - 2,404}{87.3} \right) \right] \quad (30)$$

Although the electrolyte measured in Refs. [38] and [39] contains only ions Li⁺, S₆²⁻, and A⁻, we assume that the expression derived above for the mobility of Li⁺ remains the same when other

LiPS species are used. To compute the mobility of the two anions we need to make some assumptions about their transference numbers. The easiest assumption is to assume that the transference numbers of anions are equal to each other and use the constraint that $\sum_{i=1}^n t_i = 1$, where $n = 8$ is the total number of species i in the electrolyte. However, this assumption results in $u_i = \frac{\kappa(c_{Li})}{z_i^2 F c_i} t_i = \frac{\kappa(c_{Li})}{z_i^2 F c_i} \frac{1 - t_+(c_{Li})}{n - 1}$, which implies that $u_i \rightarrow \infty$ when $c_i \rightarrow 0$, for $i \neq \text{Li}^+$. To avoid this singularity, which contradicts the diluted solution theory, we need to assume that the transference numbers of anions depend linearly on the concentration of each ion, which is also suggested by eq. (27). The easiest way is to assume that they depend linearly on each ion concentration $t_i = \frac{-z_i c_i}{c_{Li}} [1 - t_+(c_{Li})]$ for all $i \neq \text{Li}^+$. In this way, we arrive at the following equation for the mobilities of anions in the electrolyte

$$\begin{aligned} u_i &= \frac{\kappa(c_{Li})}{z_i^2 F c_i} t_i = \frac{\kappa(c_{Li})}{z_i^2 F c_i} \times \frac{-z_i c_i}{c_{Li}} [1 - t_+(c_{Li})] \\ &= \frac{\kappa(c_{Li})}{|z_i| F c_{Li}} [1 - t_+(c_{Li})] \quad \text{for all } i \neq \text{Li}^+ \end{aligned} \quad (31)$$

Introducing (29) into (31) we obtain the following equation for the mobility of the anions

$$\begin{aligned} u_i &= \frac{3.123 \times 10^{-9}}{|z_i|} \times e^{-\frac{c_{Li}}{1074}} \left[3.19 - \tanh \left(\frac{c_{Li} - 2,404}{87.3} \right) \right] \\ &\times \quad \text{for all } i \neq \text{Li}^+ \end{aligned} \quad (32)$$

The diffusivities of the Li-S catholytes should be computed using the Einstein relations $D_i = \left(1 + \frac{d \ln f_i}{d \ln c_i} \right) V_T u_i$, where f_i are the molar concentration activity coefficients and $V_T = \frac{kT}{q}$ is the thermal voltage. Due to the lack of available data in the literature for the activity coefficients in Li-S catholytes, we approximate, what is usually done in the literature

$$D_i \approx V_T u_i \quad (33)$$

However, notice that according to measurements of the activity

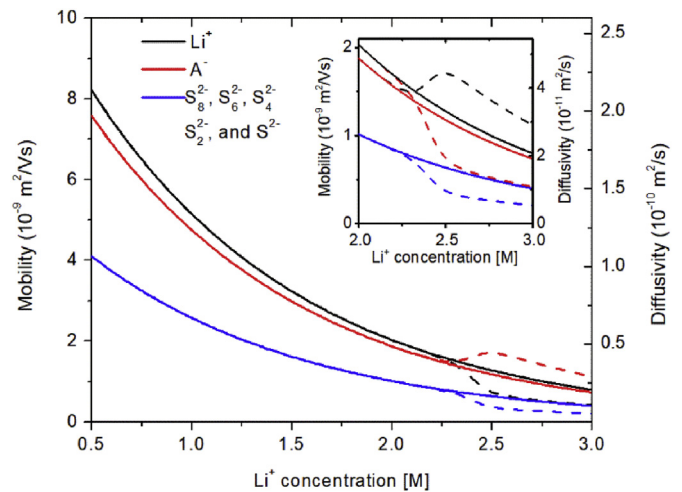


Fig. 6. Ion mobilities and diffusivities computed using eqs. (30), (32) and (33). The continuous lines are computed using a constant transference number of Li⁺ of 0.48. The dashed lines are computed using eqs. (29) and (32). The inflection points at $c_{Li} \approx 2.4$ M are due to the sudden change of the Li⁺ transference number at this concentration.

coefficients in LiPF₆-based electrolytes made by Valoen et al. [42] we expect eq. (33) to slightly underestimate the value of the diffusion coefficients.

Eqs. (30), (32) and (33) describe the mobilities and diffusivities of the ions in the electrolyte. The coefficients in these equations have been identified for LiTFSI to DME:DOL (which is the electrolyte used in our experiments) and they need to be slightly adjusted when used for other Li-S electrolytes, such as LiTFSI in diglyme, triglyme or tetraglyme. By allowing the mobilities and diffusivities of the ions to depend on the concentration of the electrolyte we indirectly assume that the ion mobilities and diffusivities depend on the state-of-charge of the cell and effects such as electrolyte viscosity are included in simulations.

Fig. 6 presents the dependence of the ion mobilities and diffusivities as a function of the lithium concentration. The dashed lines are computed using eqs. (29) and (32) and the continuous lines are computed using a constant transference number of Li⁺ of 0.48. The diffusivities are computed using eq. (33). Note that the values of the diffusivities are relatively close to the values assumed in Refs. [11,15] but significantly higher than the values used in Ref. [12]. Due to the high values of the diffusion coefficients of ionic species in LiTFSI to DME:DOL, the rate-dependent capacities cannot be attributed to the sluggish ion transport in batteries using this electrolyte. Relatively similar values for the transference number of Li⁺ and diffusion coefficients of Li⁺, A[−], and solvent molecules in LiTFSI have been recently measured experimentally by Hayamizu [43] using pulsed-gradient spin-echo nuclear magnetic resonance (NMR). For instance, the transference number of Li⁺ was approximately 0.41, while the diffusion coefficients of Li⁺ and A[−] were between 6×10^{-10} and 9×10^{-10} m²/s. The slightly larger values reported for the diffusivities of Li⁺ and A[−] are due to approximation (33).

Fig. 6 shows that the mobilities and diffusivities of polysulfides decrease significantly when the concentration of Li ions is increasing. This effect is expected because polysulfides tend to form aggregates leading to a gel-type consistency of the electrolyte when their concentration, particularly the concentration of higher-order species, is increased. Although eqs. (30) and (32) depend mainly on the concentration of Li-ions and do not depend explicitly on the concentration of each anion, these equations are still expected to describe the degradation of mass transport properties correctly under normal Li ion and polysulfide concentrations.

3.4. Solubility products

The solubility products of solid species in the electrolyte play a major role on the specific capacity of Li-S batteries [7,29,44]. Therefore, we need to carefully determine the solubility products of solid species in the cathode. In the previous reported simulations the solubility products were not determined experimentally and their values were simply assumed.

The solubility of elemental sulfur changes significantly with the concentration of the LiPS in the electrolyte. For instance, the solubility of sulfur in DME is 9.96 mol/m³, while the solubility in an electrolyte from discharged Li-S batteries is 16.6 mol/m³ [45]. In our simulations, due to the lack of available data in the literature, we use $K_{S_8} = 10$ mol/m³. This value is approximately half of the currently assumed value of 19 mol/m³ used in most numerical simulations [11,12,15].

To evaluate the solubility of LiPS we mix Li₂S and S powder in stoichiometric ratios of 1: *n*-1 to produce Li₂S_{*n*} in a blank electrolyte of DME:DOL (1:1 v:v). The solubility products of Li₂S_{*n*} is measured separately for *n* = 4, 6, and 8 by slowly increasing the amount of blank electrolyte until the powder is fully dissolved. For more

details on how the experiments were performed we recommend [7]. The measured concentrations of saturation for Li₂S₄, Li₂S₆, Li₂S₈ are [S] = 2 M, 6 M, and 7 M, respectively, which lead to the solubility products presented in Table 1. These values are in agreement with data published by other authors for similar electrolytes. For instance, Pan et al. [46] found that the maximum sulfur concentration of the Li₂S₆ in DME:DOL is $\sim$ [S] = 6 M.

The solubilities of Li₂S and Li₂S₂ are negligible according to our experiments and other published data [11,12,15,27,47–49].

3.5. Exchange current densities and reaction potentials

The voltage drop in Li-S batteries is due in part to diffusion and conductivity losses and in part to kinetic losses. Since the diffusion coefficients and mobilities have already been determined in section 3.2 independently (i.e. not by fitting the discharge curves), the diffusion and conductivity losses are accurately accounted in our simulations and the remaining voltage drop is due entirely to kinetic losses. Therefore, we can determine the exchange current densities and reaction potentials that enter in the Butler-Volmer equation by fitting plateaus of the discharge voltage and the slope between two plateaus with the numerical simulations. The values of the exchange current densities and reaction potentials determined in this way are presented in Table 2, while the reference concentrations are presented in Table 3.

3.6. Surface oversaturation

The main effect of parameters λ_i is to decrease the specific capacity of the battery as a function of the reaction rate. Therefore, parameters λ_i are determined by fitting the total specific capacity of the battery at different discharge rates with simulation results. To simplify the identification but also the numerical implementation we focus only on the formation of Li₂S_(s) nuclei and set the values of λ_i for all the other products to zero. The results of this identification are presented in Table 3.

3.7. Nucleation related parameters

The parameters describing the nucleation of solid species are K'_k and M_k . Since these parameters appear as product in surface coverage for precipitate *k* (which can be observed by solving eqs. (17)–(19)), they cannot be determined simultaneously from the surface coverage. Therefore, in this work we set $K'_k = 10^2$ m^{−2} s^{−1} and determine M_k by fitting the experimental discharge characteristics with simulations results. Due to the large number of fitting parameters and to the fact that the existing experimental reports do not suggest unanimously that other reaction products besides Li₂S are present in the battery, we use negligibly small values for $M_{Li_2S_{8(s)}}$, $M_{Li_2S_{6(s)}}$, $M_{Li_2S_{4(s)}}$, and $M_{Li_2S_{2(s)}}$ and focus only on fitting $M_{Li_2S_{(s)}}$. The results of this identification are presented in Table 1.

4. Results and discussion

The transport model presented above was implemented in our in-house simulator, RandFlux [50], which is a finite element simulation platform that allows solving general systems of one-dimensional (1-D), two-dimensional (2-D), and three-dimensional (3-D) partial differential equations. Currently, the platform implements defaults models for the simulation of electrochemical systems described by the multicomponent and concentrated solution theory (e.g. Li-S, metal-air, and Li-ion batteries), fuels cells, nanoscale and power semiconductor devices. The discretized transport systems of equations are solved using the

Table 1
Parameters for dissolution, nucleation, and precipitation.

Product (k, k')	Initial fraction, $\varepsilon_{k', initial}$		Rate constant, $k_{k'}$	Solubility product, $K_{sp, k'}$	Molar volume, $V_{k'}$ ($\text{m}^3 \text{mol}^{-1}$)	$K'_k (\text{m}^2)$	$M_k (\text{m}^{-2} \text{s}^{-1})$	N_k
	Separator	Cathode						
$\text{S}_{8(s)}$	0	0.087	1 s^{-1}	10 mol/m^3	1.239×10^{-4}			
$\text{Li}_2\text{S}_{8(s)}$	10^{-6}	10^{-6}	$10^3 \text{ mol}^{-2} \text{m}^6 \text{s}^{-1a}$	$1.531 \times 10^9 \text{ mol}^3 \text{m}^{-9}$	1.361×10^{-4}	10^2	10^{-26a}	1
$\text{Li}_2\text{S}_{6(s)}$	10^{-6}	10^{-6}	$10^3 \text{ mol}^{-2} \text{m}^6 \text{s}^{-1a}$	$2 \times 10^9 \text{ mol}^3 \text{m}^{-9}$	1.038×10^{-4}	10^2	10^{-26a}	1
$\text{Li}_2\text{S}_{4(s)}$	10^{-6}	10^{-6}	$10^3 \text{ mol}^{-2} \text{m}^6 \text{s}^{-1a}$	$5 \times 10^8 \text{ mol}^3 \text{m}^{-9}$	7.415×10^{-5}	10^2	10^{-26a}	1
$\text{Li}_2\text{S}_{2(s)}$	10^{-6}	10^{-6}	$10^3 \text{ mol}^{-2} \text{m}^6 \text{s}^{-1a}$	$3 \times 10^3 \text{ mol}^3 \text{m}^{-9}$	4.317×10^{-5}	10^2	10^{-26a}	1
$\text{Li}_2\text{S}_{(s)}$	10^{-5}	10^{-5}	$5 \times 10^8 \text{ mol}^{-2} \text{m}^6 \text{s}^{-1}$	$3 \times 10^3 \text{ mol}^3 \text{m}^{-9}$	2.768×10^{-5}	10^2	1.5×10^{-21}	1.2

^a These values are negligibly small and do so significantly influence the solid precipitation.

Table 2
Kinetic parameters at cathode (calculated by fitting the discharge curves). The exchange current density at the anode is $i_{0, \text{Li}^+} = 1 \text{ A/m}^2$.

Reaction (j)	$i_{0,j}$ (A m^{-2})	$U_{j, \text{ref}}$ (V)
(2)	0.6	2.40
(3)	0.1	2.39
(4)	10^{-2}	2.18
(5)	10^{-4}	1.95
(6)	10^{-8}	1.95

Table 3
Reference concentrations (calculated based on the assumed values of $U_{j, \text{ref}}$) and parameters λ_i that define the surface oversaturation.

Species (i)	z_i	$c_{i, \text{ref}}$ (mol m^{-3})	λ_i (s m^{-1})
Li^+	1	1	0
S_8	0	1.90×10^{-2}	0
S_8^{2-}	-2	3.97×10^{-4}	0
S_6^{2-}	-2	8.88×10^{-4}	0
S_4^{2-}	-2	8.69×10^{-6}	0
S_2^{2-}	-2	1.17×10^{-11}	0
S^{2-}	-2	1.36×10^{-14}	10^{10}
A^-	-1	0.9974 ^a	^b

^a Computed based on charge neutrality.

^b Species not participating in the electrochemical reaction.

Newton method, in which the Jacobians are computed by automatic differentiation.

For the battery simulated in this work, the initial $\text{S}_{8(s)}$ covers most of the active area of the CNTs (see Fig. 1b), which results in a relatively small dip at the beginning of the discharge curves for high discharge rates. Therefore, we model the specific area that enters in the left-hand side of eq. (20) by an empirical function

$$a = a_0 \exp \left[\left(\frac{\varepsilon_{\text{S}_{8(s)}}}{\varepsilon_{\text{S}_{8(s), \text{initial}}}} \right)^m \ln \theta_{\text{ini}} \right] \quad (34)$$

where m is a fitting parameter, a_0 is the total specific surface area of the carbon, and θ_{ini} is the initial fraction of the uncovered area of the cathode. In our simulations $\theta_{\text{ini}} = 10^{-3}$, which means that 99.9% of the cathode surface is initially covered with $\text{S}_{8(s)}$. The total specific surface area of the carbon was estimated from the total surface area ($392 \text{ m}^2/\text{g}$) and the mass density of the CNT foam ($1.25 \times 10^5 \text{ g/m}^3$) to be $a_0 = 4.9 \times 10^7 \text{ m}^2/\text{m}^3$. The values of all the other geometry and structure related parameters of the battery simulated in this section are presented in Table 4.

Eqs. (17)–(22) were discretized on a 1-D grid that covered the separator and cathode regions. The transport system was subject to appropriate boundary conditions (see for instance [11]) and solved iteratively in time using the backward Euler method. The total

Table 4
Miscellaneous model and geometry parameters.

Parameter	Value
α	0.5
a_0	$4.9 \times 10^7 \text{ m}^2/\text{m}^3$
b	1.5
L_C	10^{-4} m
L_S	$2.5 \times 10^{-5} \text{ m}$
m	1.2
ε_{CNT} (cathode)	0.06
ε (separator)	0.41
V_A	0 V
V_T	0.026 V
U_0	2.45 V
σ	10 S/m
θ_{ini}	10^{-3}

number of mesh points was 100, which proved to be sufficiently high to minimize the discretization errors. The total time for the computation of the discharge voltage of the battery at a given discharge current is less than five minutes on a personal computer operating at 3 GHz.

The simulated voltage characteristics under different discharge rates ranging from 0.1 C to 1 C are represented in Fig. 7. Notice that the model describes correctly the most important features of the voltage characteristics, including the two voltage dips observed in the experimental characteristics (see Fig. 2) and the specific capacity of the cells as a function of the discharge rates. The slight

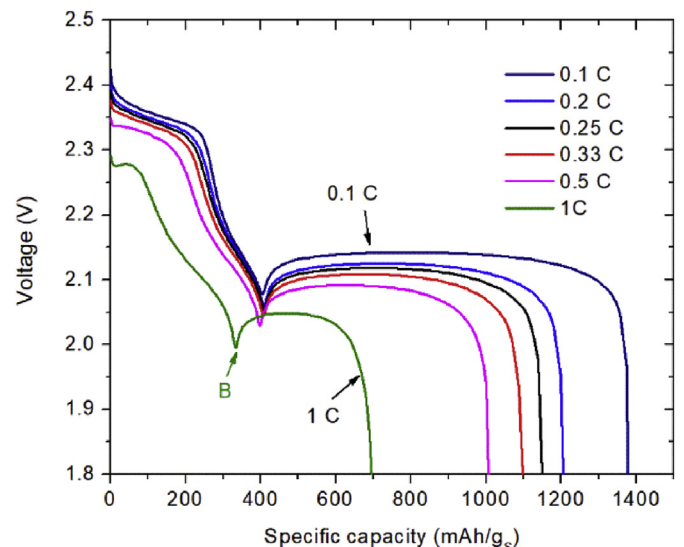


Fig. 7. Simulated cell voltages under different discharge rates.

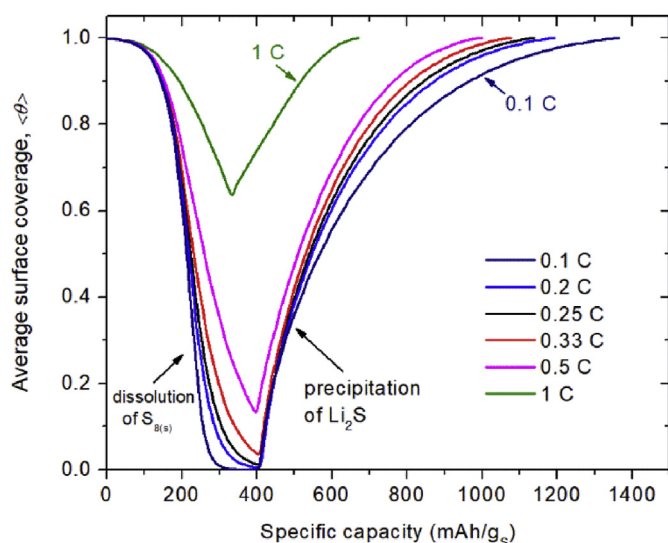


Fig. 8. Average surface coverage in the cathode under different discharge rates.

differences between the experimental and simulated discharge characteristics at 1 C during the first part of the discharge are due to the simplistic way in which eq. (34) treats surface coverage by solid sulfur.

The main reason for the decrease of the specific capacity at high discharge rates is cathode passivation due to deposition of solid products. This fact, can be observed in Fig. 8, which presents the average surface coverage, $\langle \theta \rangle$, under different discharge rates. The average surface coverage is computed by averaging the surface coverage, $\theta(\mathbf{r})$, at each mesh point inside the cathode, according to $\langle \theta \rangle = \frac{1}{V_{\text{cathode}}} \int_{\text{cathode}} \theta(\mathbf{r}) d\mathbf{r}$, where V_{cathode} is the volume of the cathode. During the first part of the discharge (i.e. before reaching the oversaturation point) the surface coverage decreases because of the dissolution of solid sulfur, which was initially deposited on the surface of the carbon. During the second part of the discharge, the surface coverage increases because of the production and deposition of reaction products $\text{Li}_2\text{S}_{8(s)}$, $\text{Li}_2\text{S}_{6(s)}$, $\text{Li}_2\text{S}_{4(s)}$, $\text{Li}_2\text{S}_{2(s)}$, and $\text{Li}_2\text{S}_{(s)}$. As expected, the battery cannot be further discharged when the surface coverage becomes equal to 1. This is the main reason for the sudden death of the Li-S batteries in our study.

Fig. 8 also shows that, at high discharge rates, the solid deposition of Li_2S can start before the entire solid $\text{S}_{8(s)}$ has been dissolved in the electrolyte. For this reason, if the deposition of Li_2S occurs on the surface of $\text{S}_{8(s)}$ it is possible that some sulfur might remain isolated and not able to participate in the electrochemical reactions. This fact can be another reason why the Li_2S batteries fabricated in the laboratory can hardly reach their maximum theoretical discharge capacity. In order to avoid this phenomenon and improve the specific capacity of Li-S batteries one should use electrolytes with high solubility of $\text{S}_{8(s)}$.

Fig. 9 presents the average number of nuclei of Li_2S per unit area on the carbon, $F_{\text{Li}_2\text{S}}$, as a function of specific capacity. In agreement with the discharge characteristics, we observe three main regions on this figure. “Region 1” corresponds to the initial dissolution of solid sulfur in the cathode. In this region, the number of nuclei per unit area is increasing slowly but still remains relatively negligible. In “region 2”, the concentration of Li_2S molecules increases abruptly, oversaturating the surface of the carbon. The nucleation rate is very high in this region. When enough nuclei have been formed and they provide enough growing seeds for the solid product deposition, the concentration of Li_2S starts decreasing and

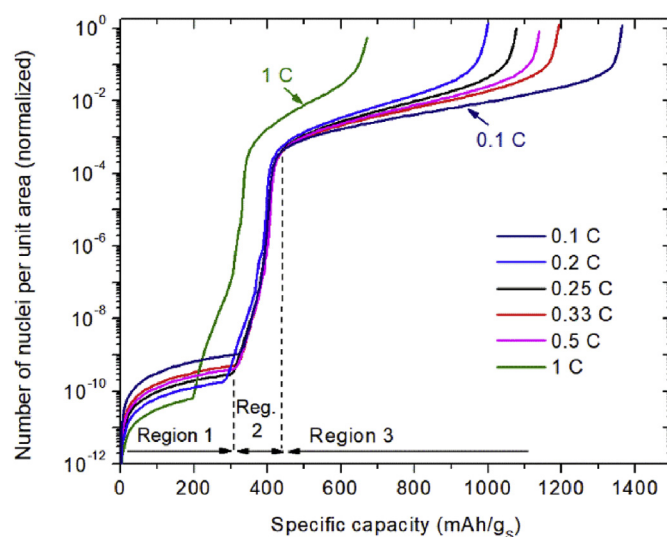


Fig. 9. Normalized average number of Li_2S nuclei per unit area.

the solid product starts growing on the surface of the cathode. This growing process takes place in “region 3”. The oversaturation dip labeled with letter “B” in Figs. 2 and 7 takes place at the end of “region 2” and beginning of “region 3”.

To emphasize the importance of using the surface oversaturation during the nucleation process, Fig. 10 presents the average oversaturation in the transport channels ($S_{\text{Li}_2\text{S},\text{surf}}$, see eq. (16)) and on the surface of the carbon ($S_{\text{Li}_2\text{S},\text{surf}}$, see eq. (15)) at the same discharge rates as in the previous two figures. During the first part of the discharge (i.e. on the first plateau) both $S_{\text{Li}_2\text{S}}$ and $S_{\text{Li}_2\text{S},\text{surf}}$ are equal to zero because the nucleation seeds have not been formed yet. Starting at approximately 200 mAh/g_s for the discharge at 1 C and at around 300 mAh/g_s for the discharge at the other rates, both $S_{\text{Li}_2\text{S}}$ and $S_{\text{Li}_2\text{S},\text{surf}}$ increase and the nucleation seeds start to form on the surface of the carbon. Once the nuclei are formed, the oversaturation $S_{\text{Li}_2\text{S}}$ starts decreasing monotonically (see Fig. 10a), however the surface oversaturation $S_{\text{Li}_2\text{S},\text{surf}}$ increases further (see Fig. 10b). This behavior is due to the fact that, when the active area of the cathode is decreasing, the reaction rate given by eq. (23) needs to increase correspondingly to provide the same amount of discharge current. Consequently, the nucleation rate, which is proportional to the surface oversaturation, increases even more during the second plateau of discharge.

It is apparent from Fig. 10 that the electrolyte needs to be significantly oversaturated in order for the nucleation process to start and the battery to be discharged at a rate of 0.1 C or above. From eq. (24) and the values of the solubility products and reaction rates reported in Table 3 it follows that the dissolution of solid species is a very slow process (at least a few orders of magnitude slower than the precipitation rate under normal discharge conditions). For this reason, in order to improve the specific capacity of Li-S batteries under normal discharge conditions it is essential to avoid the solid formation of intermediate LiPS and the only solid material formed in the cathode to be the final Li_2S product. This effect could be achieved by increasing the solubility of the intermediate LiPS as much as possible and protecting the anode electrode. A similar result was obtained experimentally by Shen et al. [7] in Li-S batteries.

Next, let us focus on the discharge characteristics at variable rate. By following the same procedure as in the experimental section, we discharge the battery by approximately 50% at 0.2 C, then

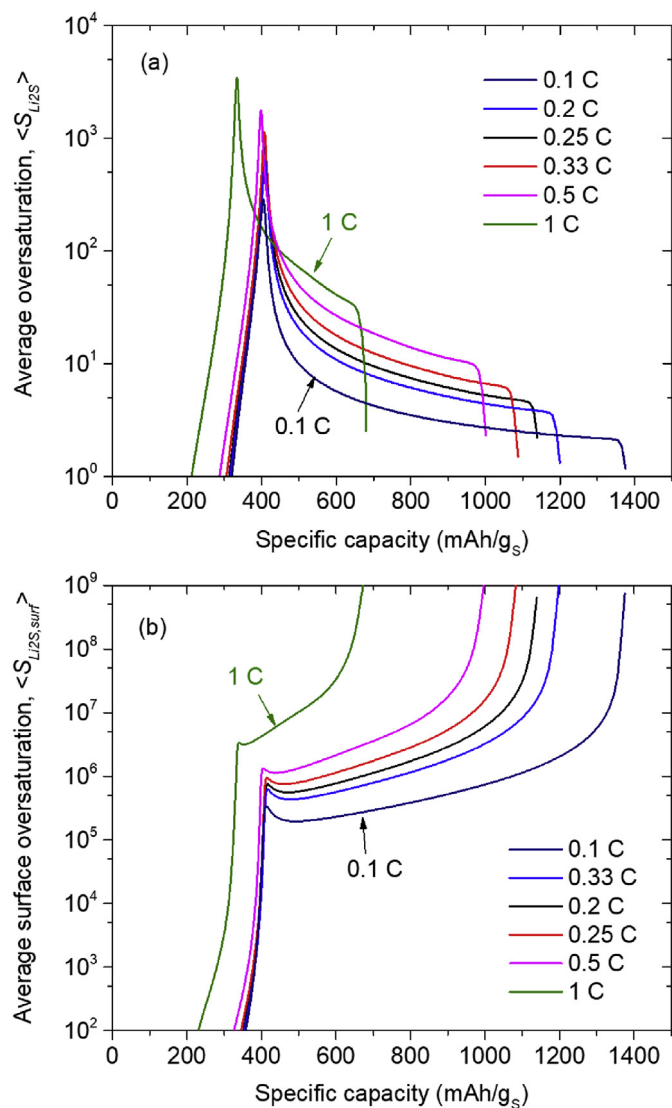


Fig. 10. Average oversaturation (a) and surface oversaturation of Li_2S (b) in the cathode as a function of the specific capacity.

we discharge at 1 C until the voltage of the battery is 1.8 V (i.e. the battery is almost dead), after which we continue with a discharge at 0.2 C. The result of this simulation is presented in Fig. 10 with continuous, while the discharges at 1 C and 0.2 C are represented with dash lines. The two main observations made in section 2 about the experimental voltage curve at variable discharge rate are also valid for the simulations presented in Fig. 11.

First, in qualitative agreement with the experimental data shown in Fig. 3, the voltage of the battery during the discharge from point “F” to point “G” is smaller than the voltage between the same specific capacities on the 0.2 C curve (see the red dash line in Fig. 11). In addition, the total specific capacity obtained by discharging the battery at variable rate (at point “G”) is smaller than the specific capacity obtained by discharging the battery at a constant rate of 0.2 C. These results are due to the fact that the rate at which the nuclei appear during the 1 C region of the discharge are different from the rate at which they are would appear if the rate was 0.2 C; for this reason, the rate at which surface coverage changes is different during the 1 C discharge (i.e. from point “E” to “F”) than during discharge at 0.2 C, and the cathode becomes passivated faster at 1 C than at 0.2 C.

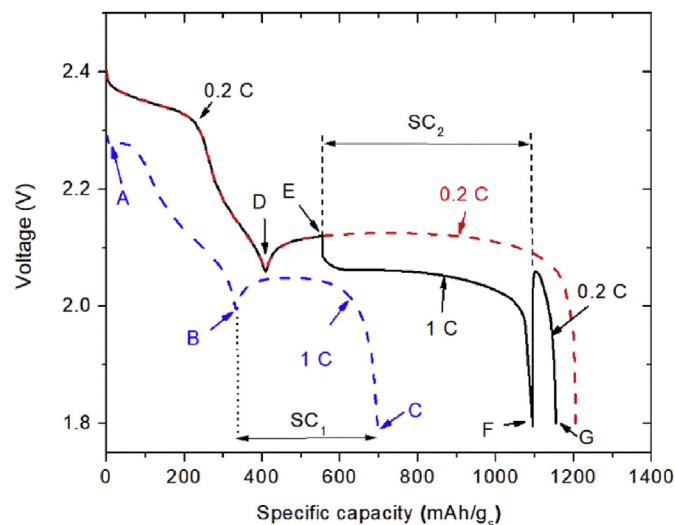


Fig. 11. Simulated cell voltage under variable discharge rate of consecutive 0.2 C-1 C-0.2 C steps (continuous line). The dash lines represent the cell voltage under constant discharge rates of 0.2 C and 1 C, respectively. Specific capacity SC_1 measured from the beginning of the precipitation process until the death of the battery at 1 C is smaller than specific capacity SC_2 .

Second, also in agreement with the experimental data, the capacity from point “B” to “C” in Fig. 11 (SC_1) is smaller than the capacity from point “E” to “F” (SC_2). This result is somewhat unexpected because the cathode is partly covered with solid Li_2S at point “E” and we expect that the remaining capacity be smaller than the one obtained by starting from point “B”, where the cathode is completely uncovered. The explanation of this phenomenon is due to the fact that the surface coverage increases much faster from point “B” to “C” than from “E” to “F”. Indeed, in the case of constant discharge at 1 C, the initial number of nuclei at point “B” is 0, which results in a large surface oversaturation and a formation of a large number of nucleation seeds at point “B”. These seeds will grow in size uniformly from “B” to “C” until completely covering and passivating the surface of the cathode. If the battery is discharged at 0.2 C a smaller number of nucleation seeds are formed at point “D” because the surface oversaturation is much smaller than at point “B”. The relatively small number of nuclei will grow slowly from “D” to “E”. When the discharge rate is changed to 1 C at point “E”, the cathode surface already contains a small number of nucleation seeds that provide growing centers for further discharge (from “E” to “F”). This phenomenon is consistent with the experimental results published in Fig. 3.

Finally, the above result prompts up to the conclusion that in order to increase the capacity of Li-S batteries it is better to artificially introduce a small number of nucleation seeds in the cathode, which will avoid reaching large surface oversaturations during the discharge of the battery. The number of nucleation seeds should not be too large so the nuclei will touch each other and passivate the surface of the carbon too soon during the growth process, however it should be large enough to allow the molecules of the solid product to deposit easily and avoid excessive oversaturation. Practically, this can be achieved by adding catalyst particles that would promote the nucleation of the solid product in specific locations of the cathode. The catalyst particles should work as nucleation seeds during the precipitation process. In addition to increasing the specific capacity, the catalyst particles will also eliminate the voltage dip at the beginning of the last plateau in Figs. 3 and 7.

5. Conclusions

After reviewing the various phenomena that result in rate-dependent specific capacities in Li-S batteries, we conclude that the sudden death of the batteries fabricated in our study is due to surface passivation of the active area of the cathode with solid reaction products. To explain the discharge characteristics of the Li-S batteries we developed a mathematical model based on the classical Kolmogorov-Avrami nucleation theory. However, instead of expressing the model in integral form and keeping track of the number and size distribution of the nucleation seeds at each location inside the battery as it is usually done in the literature, we write the surface coverage in the form of a system of differential equations (see eqs. (17)–(19)). This system can be coupled with the classical transport equations in multicomponent electrolytes (see eqs. (20)–(24)) and solved efficiently using the finite element technique.

An important feature of our model is that it uses surface oversaturation as the driving force for nucleation. Since the surface oversaturation is rate-dependent, the nucleation rate depends on the current rate of discharge. We have shown that, the nucleation rate is higher at high discharge rates, which results in a larger number of nuclei of relatively small size on the surface of the cathode. At low discharge rates the nucleation rate is lower, the number of nuclei is smaller, however, the nuclei grow in larger sizes. Based on this model we are able to describe qualitatively correctly the decrease of the capacity of Li-S batteries with increasing discharge rate and explain the experimental values of the specific capacity obtained under variable discharge currents.

The model can also be applied to the study of electrochemical impedance spectra (EIS) of Li-S batteries by linearizing the transport equation and applying the small-signal perturbation technique to evaluate the complex impedance. EIS can provide important information about the kinetic dynamics and relaxation times in the battery [51–53].

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Appendix A. Equation Chapter (Next) Section 1

To develop a model for the growth of solid products we assume that nucleation takes place on the surface of the electron conductive material and nuclei grow in the perpendicular direction to the interface and parallel to the surface (see Fig. A.1). We also assume that the precipitation process starts at $t = 0$ and nuclei are formed on the surface of the electron conductive material with a time-dependent nucleation rate $n_k(t)$ (expressed in nuclei per unit area and unit time), where index k can be $\text{Li}_2\text{S}_{8(s)}$, $\text{Li}_2\text{S}_{6(s)}$, $\text{Li}_2\text{S}_{4(s)}$,

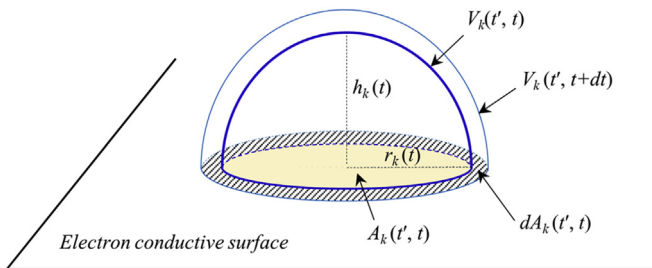


Fig. A.1. Growth of the precipitate on the electron conductive surface (i.e. carbon).

$\text{Li}_2\text{S}_{2(s)}$, and $\text{Li}_2\text{S}_{(s)}$. We denote the volume at time t of one nucleus formed at time t' by $V_k(t', t)$ and the surface area occupied by this nucleus by $A_k(t', t)$, $t' \leq t$. V_k and A_k are the volume and the coverage calculated by assuming that the nucleus does not touch or interact with other nuclei during the growth process.

The driving force for the grown of the nuclei is oversaturation, $S_k(t)$. If $\sigma_k(t', t)$ is the external surface area at time t of one nucleus formed at time t' (excluding the area of the base), the rate at which the volume of the nucleus changes in time is

$$\frac{\partial V_k(t', t)}{\partial t} = K_k \sigma_k(t', t) S_k(t), t \geq t' \quad (\text{A.1})$$

where $V_k(t', t') = 0$ and K_k is a rate constant describing how fast the nucleus grows once the seed is formed. To find the dependence of the external surface area σ_k on volume V_k we assume that the precipitates are semispherical with radius r_k . In the case of semispherical precipitates $\sigma_k = 2\pi r_k^2$, which yields $\sigma_k = (18\pi)^{1/3} V_k^{2/3}$. Using this equation, eq. (A.1) becomes

$$\frac{\partial V_k(t', t)}{\partial t} = (18\pi)^{1/3} K_k V_k^{2/3}(t', t) S_k(t), t \geq t' \quad (\text{A.2})$$

which can be integrated to give

$$V_k(t', t) = 3(18\pi) K_k^3 \left[\int_{t'}^t S_k(\tau) d\tau \right]^3 t \geq t' \quad (\text{A.3})$$

The surface on which the precipitates grow is limited by the total surface area of the electrode, $A_{\max}$. Therefore, by following the analysis by Kolmogorov [26] and Avrami [54,55], we write the rate at which the area covered by all the nuclei of species k increases in time as

$$\frac{dA_{k,\text{total}}(t)}{dt} = \left[1 - \frac{A_{\text{total}}(t)}{A_{\max}} \right] \frac{dA_{k0}(t)}{dt} \quad (\text{A.4})$$

where $A_{\text{total}}(t) = \sum_k A_{k,\text{total}}(t)$ is the total area covered by the precipitates and $\frac{dA_{k0}(t)}{dt}$ is the rate at which the surface covered by the nuclei would grow if there were no overlapping regions (i.e. all the precipitates would grow isolated from each other). Dividing eq. (A.4) by $A_{\max}$, we obtain

$$\frac{d\theta_k(t)}{dt} = [1 - \theta(t)] \frac{d\theta_{k0}(t)}{dt} \quad (\text{A.5})$$

where $\theta_k(t) = \frac{A_{k,\text{total}}(t)}{A_{\max}}$ is the surface coverage, $\theta(t) = \sum_k \theta_k(t)$, and $\theta_{k0}(t)$ is the hypothetical surface coverage calculated as if the growth fronts did not overlap

$$\theta_{k0}(t) = \int_0^t n_k(t') A_k(t', t) dt' \quad (\text{A.6})$$

Notice that by integrating (A.5) we obtain the 2-D Kolmogorov equation, $\theta_k(t) = 1 - \exp[-\int_0^t n_k(t') A_k(t', t) dt']$, which was used in Refs. [14,25] to model surface coverage in Li-air and Li-S batteries. However, in this work, we are not interested in finding an integral equation for the surface coverage as we aim at deriving a differential system of equations that describes surface coverage $\theta_k(t)$. Introducing eq. (A.6) into (A.5) and using that $A_k(t', t') = 0$, we obtain the following equation for surface coverage

$$\frac{d\theta_k(t)}{dt} = [1 - \theta(t)] \int_0^t n_k(t') \frac{dA_k(t', t)}{dt} dt' \quad (\text{A.7})$$

The coverage area of one nucleus depends on volume V_k . For semispherical nuclei $A_k = \pi r_k^2$, $V = 2\pi r_k^3/3$ and we obtain $A_k = (9\pi/4)^{1/3} V_k^{2/3}$. Hence, eq. (A.7) becomes

$$\frac{d\theta_k(t)}{dt} = \frac{2(9\pi/4)^{1/3}}{3} [1 - \theta(t)] \int_0^t n_k(t') V_k^{-1/3}(t', t) \frac{\partial V_k(t', t)}{\partial t} dt' \quad (\text{A.8})$$

Using (A.2) and (A.6), eq. (A.8) becomes

$$\frac{d\theta_k(t)}{dt} = 6\pi^{1/3} K_k^2 S_k(t) [1 - \theta(t)] \int_0^t n_k(t') \left[\int_{t'}^t S_k(\tau) d\tau \right] (t', t) dt' \quad (\text{A.9})$$

The last integro-differential equation can be written in the form of a system of first-order differential equations by introducing notations

$$F_k(t) = \int_0^t n_k(t') \left[\int_{t'}^t S_k(\tau) d\tau \right] (t', t) dt' \quad (\text{A.10})$$

$$G_k(t) = \int_0^t n_k(t') dt' \quad (\text{A.11})$$

Since

$$\begin{aligned} \frac{dF_k(t)}{dt} &= n_k(t') \left[\int_{t'}^t S_k(\tau) d\tau \right]_{t'=t} + \int_0^t n_k(t') \frac{d}{dt} \left[\int_{t'}^t S_k(\tau) d\tau \right] dt' \\ &= S_k(t) \int_0^t n_k(t') dt' = S_k(t) G_k(t) \end{aligned} \quad (\text{A.12})$$

eq. (A.9) can be split into three ordinary differential equations

$$\frac{d\theta_k(t)}{dt} = 6\pi^{1/3} K_k^2 S_k(t) [1 - \theta_k(t)] F_k(t) \quad (\text{A.13})$$

$$\frac{dF_k(t)}{dt} = S_k(t) G_k(t) \quad (\text{A.14})$$

$$\frac{dG_k(t)}{dt} = n_k(t) \quad (\text{A.15})$$

The initial conditions for functions F_k and G_k can be obtained by introducing $t = 0$ in (A.10) and (A.11), respectively: $F_k(0) = G_k(0) = 0$. By defining $K'_k = 6\pi^{1/3} K_k^2$ system (A.13)–(A.15) leads to eqs. (17)–(19).

Appendix B. Nomenclature

a	total specific area of the cathode excluding the area covered by $S_{8(s)}$, $\text{m}^2 \text{m}^{-3}$
a_0	specific area of the cathode including the area covered by $S_{8(s)}$, $\text{m}^2 \text{m}^{-3}$

$a_{i,surf}$	activities of species i at the surface of the carbon, mol m^{-3}
A^-	anions in the LiTFSI salt and LiNO_3 electrolyte additive
b	Bruggeman constant
c_i	concentration of species i in the channels, mol m^{-3}
$c_{i,surf}$	concentration of species i at the surface of the carbon, mol m^{-3}
D_i	diffusion coefficient of species i in bulk material, $\text{m}^2 \text{s}^{-1}$
D'_i	effective value of the diffusion coefficient of species i , $\text{m}^2 \text{s}^{-1}$
f_i	molar concentration activity coefficients of species i
F	Faraday constant, $96,485 \text{ C mol}^{-1}$
F_k	integration function, $\text{m}^{-2} \text{s}$
G_k	integration function describing the number of nuclei per unit area, m^{-2}
$i_{0,j}$	exchange current density for the j -th electrochemical reaction, A m^{-2}
k	Boltzmann constant, $1.38 \times 10^{-23} \text{ J K}^{-1}$
$k_{k'}$	precipitation rate constant for species k' (units depends on specie)
K'_k	parameter describing how fast precipitation product k grows once the nucleation seed is formed, m^2
$K_{sp,k'}$	solubility product for species k' (units depends on species)
L_C	cathode width, m
L_S	separator width, m
m	parameter describing the area coverage with $S_{8(s)}$
n_j	number of electrons participating in the j -th electrochemical reaction
n_k	number of nuclei of species k that appear per unit area of the carbon and unit time, $\text{m}^{-2} \text{s}^{-1}$
M_k	parameter defined in eq. (13), $\text{m}^{-2} \text{s}^{-1}$
N_k	parameter defined in eq. (13)
q	absolute value of electron charge, $1.6 \times 10^{-19} \text{ C}$
R	ideal gas constant, $8,3143 \text{ J mol}^{-1} \text{K}^{-1}$
R_A	reaction rate at the anode (production of species i at the surface of the carbon), $\text{mol m}^2 \text{s}^{-1}$
R_i	reaction rate (production of species i at the surface of the carbon), $\text{mol m}^2 \text{s}^{-1}$
$r_{k'}$	precipitation rate of species k' (dissolution is defined as $-r_{k'}$), $\text{mol m}^3 \text{s}^{-1}$
$s_{i,j}$	stoichiometric coefficients of species i in electrochemical reaction j
S_k	oversaturation of ionic species entering the formation of solid species k in the channels
$S_{k,surf}$	oversaturation of ionic species entering the formation of solid species k at the surface of the carbon
t, t'	time, s
t_i	transference number of species i
t_+	transference number of cations (same as t_{Li^+})
T	temperature, K
u_i	mobility of species i in bulk material, $\text{m}^2 \text{s}^{-1} \text{V}^{-1}$
u'_i	effective value of the mobility of species i , $\text{m}^2 \text{s}^{-1} \text{V}^{-1}$
U_0	open circuit voltage of battery at initial conditions, V
$U_{j,ref}$	reference open circuit voltage for electrochemical reaction j , V
V_T	thermal voltage, $V_T = kT/q$
V	potential of electrons, V
V_A	open circuit voltage of the anode reaction with respect to the Li electrode, V
$V_{k'}$	partial molar volumes of the precipitates, $\text{m}^3 \text{mol}^{-1}$
z_i	charge number of species i , $\text{m}^3 \text{mol}^{-1}$
Greek symbols	

α	symmetry factor
ε	fraction of the electrolyte
$\gamma_{i,k}$	number of moles of ionic species i in the solid species k
κ	electrolyte conductivity, S m^{-1}
λ_i	parameter defined in eq. (15), s m^{-1}
η	overpotential at the cathode, V
η_A	overpotential at the anode, V
σ	conductivity of the electron conductive material, S m^{-1}
ϕ	electrostatic potential of electrolyte, V
θ	total relative coverage
θ_k	relative coverage of species k
θ_{ini}	initial fraction of the uncovered area of the cathode

Subscripts and superscripts

CNT	carbon nanotube or carbon phase
i	indexes the 8 mobile species in the electrolyte Li^+ , S_8 , S_8^{2-} , S_6^{2-} , S_4^{2-} , S_2^{2-} , and A^-
j	indexes the 5 electrochemical reactions at the cathode, eqs. (2)–(6)
k	indexes the 5 precipitates $\text{Li}_2\text{S}_{8(s)}$, $\text{Li}_2\text{S}_{6(s)}$, $\text{Li}_2\text{S}_{4(s)}$, $\text{Li}_2\text{S}_{2(s)}$, and $\text{Li}_2\text{S}_{(s)}$
k'	indexes the 6 solid species $\text{S}_{8(s)}$, $\text{Li}_2\text{S}_{8(s)}$, $\text{Li}_2\text{S}_{6(s)}$, $\text{Li}_2\text{S}_{4(s)}$, $\text{Li}_2\text{S}_{2(s)}$, and $\text{Li}_2\text{S}_{(s)}$
surf	surface of carbon
ref	reference value
(s)	solid

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